Session Objectives

- Define emerging infections.
- Explore emerging viral and bacterial pathogens, their properties, pathogenesis, etc.
- Discuss the causes that lead to the emergence of these diseases and efforts to combat that.
- Discuss the challenges emerging infectious diseases present to the medical community.

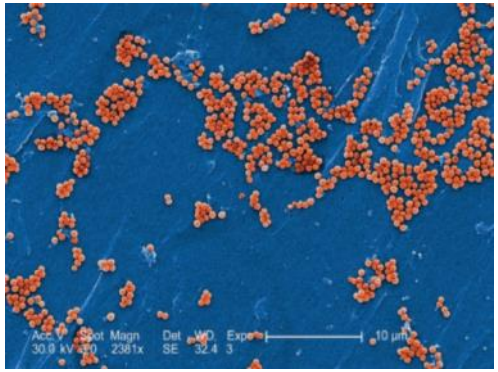
Source: Course Syllabus

EID List (WHO/CDC)

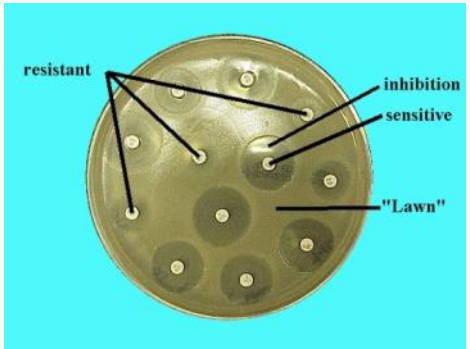
- Viral:
  - **Mumps**
  - **Measles**
  - **Influenza** (avian & swine)
  - HIV/AIDS
  - **Dengue fever**
  - **Ebola virus disease**, **Marburg** & Lassa fever
  - **MERS, SARS-CoV-2** coronavirus diseases
  - **Hantavirus pulmonary syndrome**
  - Crimean Congo hemorrhagic fever
  - Nipah & Rift Valley fever
  - Chikungunya virus disease
  - Zika virus disease
- Bacterial:
  - Antibiotic resistant bacteria: e.g. **Multidrug-resistant TB**, **Gonorrhea**, **CRE**
  - Healthcare-Associated Infections: e.g. **MRSA**, ***C. difficile***
  - **Pertussis**
  - **Cholera**
  - Anthrax
  - Tularemia

Methicillin-resistant *Staphylococcus aureus*

- Gram +ve cocci
- Normal flora of humans:
  - Nasal membranes/skin
- Opportunistic infections:
  - Mostly skin infections
  - Focal damage
    - 53% purulent wounds
    - 61% abscesses
  - Systemic disease
- Nosocomial infections
- Historically termed MRSA strains are oxacillin and methicillin resistant
- They are also resistant to all b-lactam agents, including cephalosporins & carbapenems

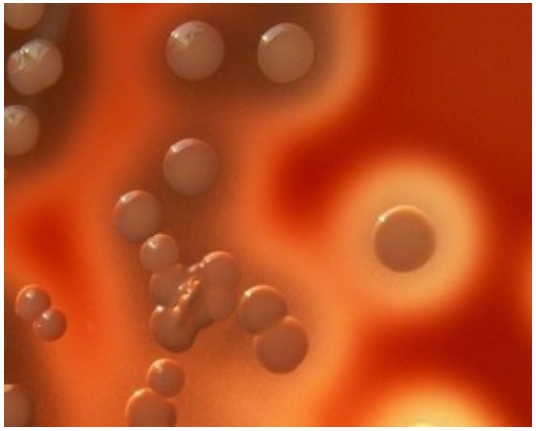






MRSA

- All members are highly resistant to drying, extremes in pH, and high temperatures (Biofilm formation)
- Increase in community acquired cases, with infections associated with schools, prisons, & shared athletic equipment
- MRSA was responsible for 94,000 life-threatening infections and nearly 19,000 deaths in the US
- VRSA



S. aureus is β -hemolytic on blood agar

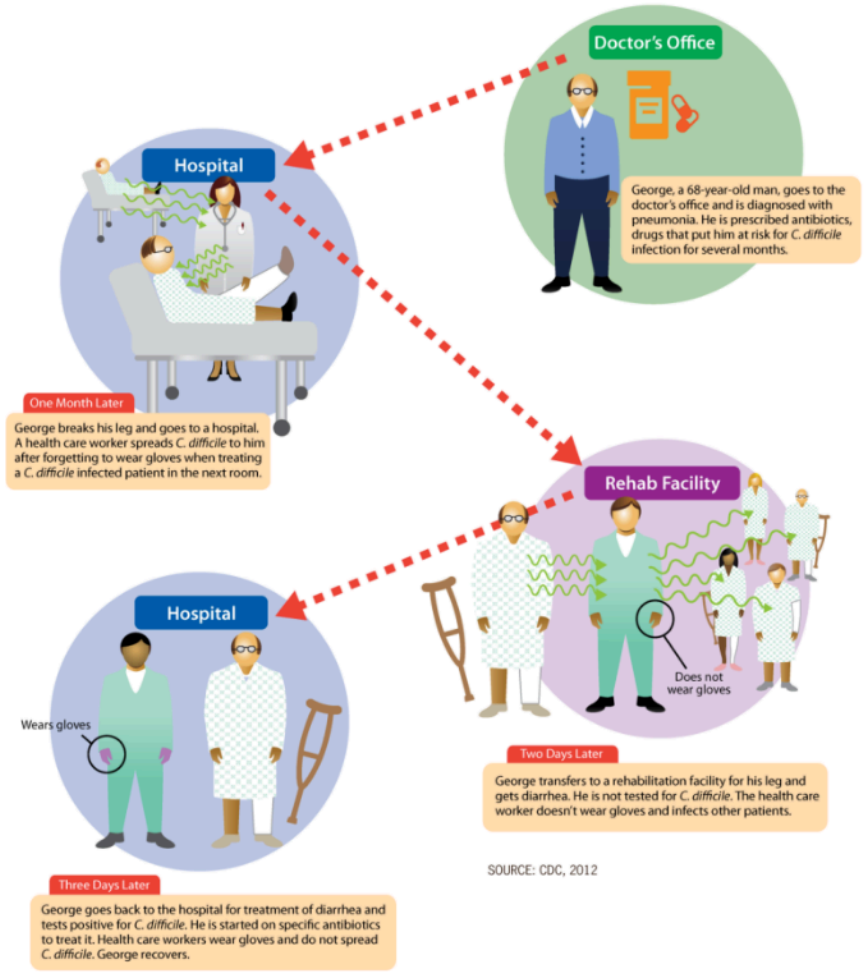
"the thoughtless person playing with penicillin is morally responsible for the death of the man who finally succumbs to infection with the penicillin-resistant organism. I hope this evil can be averted" Sir A. Fleming, 1945

The rise of *C. difficile*

Hypervirulent epidemic strain of CDI (B1/NAP1)

- Implicated in outbreaks throughout the US, Canada & Europe
- Now seen in at least 40 states, exhibit:
  - Greater toxin production
  - Greater antimicrobial resistance
- Control: infection control, environmental cleaning & antibiotic stewardship

How *C. difficile* spreads?



Multidrug-resistant TB

- Prevalent in hospitals, prisons, etc.
- HIV infected patients
- 1 in 106/108 spontaneous mutants resistant to first-line anti-TB drugs
- Combination treatment (*takes longer & require 2nd line drugs*)
- 1st line: INH, RMP, PZA & EMB
- 2nd line: kanamycin, ethionamide, cycloserine, ofloxacin & ciprofloxacin
- Resistance mechanisms identified -> mutations in target proteins
- Multidrug-resistant *M. tuberculosis*: INH & RMP

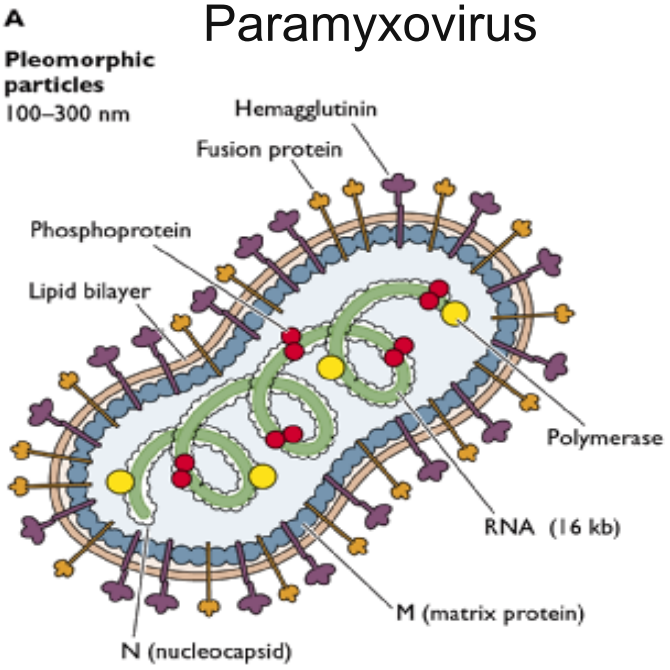
Carbapenem-resistant *Enterobacteriaceae*

- Some CRE have become resistant to most available antibiotics
- Mortality up to 50%
- Contact precautions, alert facility, infection control & laboratory staff
- Tx:
  - Best: combination therapy with two or more drugs including at least one carbapenem (carbapenem's MIC is ≤ 4 mg/L)
  - Last resort (old drugs): polymyxins (renal toxicity), fosfomycin

Measles (Rubeola)

(-) ssRNA
Worldwide
- Leading cause of child death
  - 164,000 deaths in 2008
- Measles Initiative
  - Decrease deaths due to measles
  - American Red Cross, United Nations Foundation, CDC, UNICEF, World Health Organization
  - Began in 2001 because 733,000 worldwide deaths
  - 78% drop in number of deaths by 2008

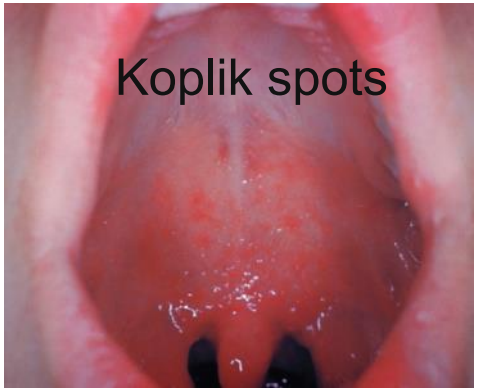
Paramyxovirus



Measles

Spread through respiratory secretions
- Coughs, sneezes
- Virus survives on surfaces up to 2 hours
- Dense rash on face, trunk, arms, legs
- Sore throat, high fever (up to 41°C), cough, runny nose, red eyes, light sensitivity
- Signs/symptoms occur 7–18 days after infection
- **Koplik spots** seen in mouth
  - Red spots with bluish white centers





Measles: Diagnosis & Treatment

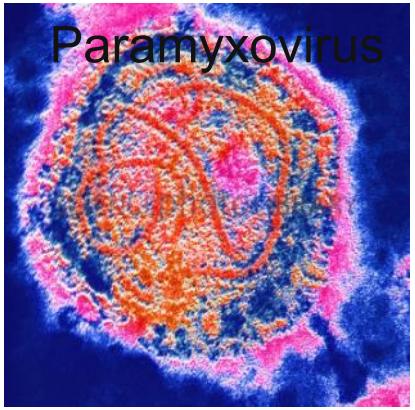
- Diagnosis
  - Test for measles Abs to determine immune status
  - Clinical diagnosis
    - Characteristic rash & fever
  - Only one antigenic strain of measles
    - Several strains can be differentiated genetically
  - Viral culture
    - Nasal or throat swab
  - RT-PCR
- Treatment
  - Acetaminophen or ibuprofen (fever)
  - No tx to decrease the viral load

Mumps

(-) ssRNA

Signs & symptoms

- 15–20% are asymptomatic
- Swollen parotid salivary glands
- Submaxillary, sublingual salivary glands may be enlarged
- Fever, headache, muscle aches, loss of appetite, fatigue





(a)



(b)

Mumps

- Spread
  - Direct contact with secretions
  - Indirect by contaminated items (fomites)
- Contagious
  - 1–2 days before swelling; up to 5 days after swelling
- Lab analyses:
  - Viral culture
  - RT-PCR
- Lab to detect IR
  - Anti-mumps IgM
  - 4-fold rise in IgG
- Mumps, rubella, & measles are reportable

African hemorrhagic fevers

- 2014 - Ebola largest outbreak
  - Guinea, Liberia & Sierra Leone
  - 10,000 death
  - Several imported cases
  - 1 in the US (exposed some flawed control measures)
- Abs protective
- Detection: ELISA & RT-PCR (special care with handling sera)
- Person to person spread (bodily fluids/blood)
- Patients isolated
- Tx: Providing fluids and electrolytes (body salts) through infusion into the vein (intravenously).
  - Offering O2 therapy
  - Using medication to support bp, reduce vomiting & diarrhea
  - Manage fever & pain
  - Treating other infections, if they occur
- No approved antiviral treatment (Regeneron (REGN-EB3) & mAb114) or vaccine (experimental)

ere acute respiratory syndrome (SARS)

- Outbreak: November 2002, highly contagious & severe atypical pneumonia were observed in the Guangdong Province of southern China
- Virus spread to Hong Kong in February 2003 by a physician who died 10 days after admission into local hospital -> a mysterious death
- Similar outbreaks occurred at different local communities subsequently
- The virus was identified by Hong Kong, the U.S. & Germany in March 2003
  - >8,450 cases, 810 death, 30% healthcare workers, 32 countries



Natural hosts of SARS

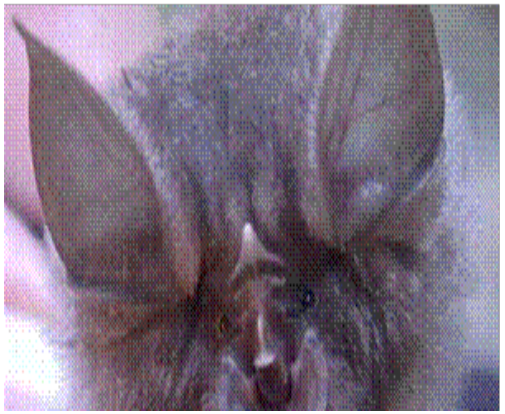
SARS may have originated in wild horseshoe bats in China (reservoir)

Indications:

- 92% similar to the human SARS virus from 3 species of bats found at the Zoology at the Chinese Academy of Sciences, Beijing
- Infected bats shows little or no symptoms

Direct human infection is not likely to occur

- The differences is in the genes which able the binding of virus particle to host cell
- Virus may jump to an intermediate animal such as the civet cat & adapt to infect humans



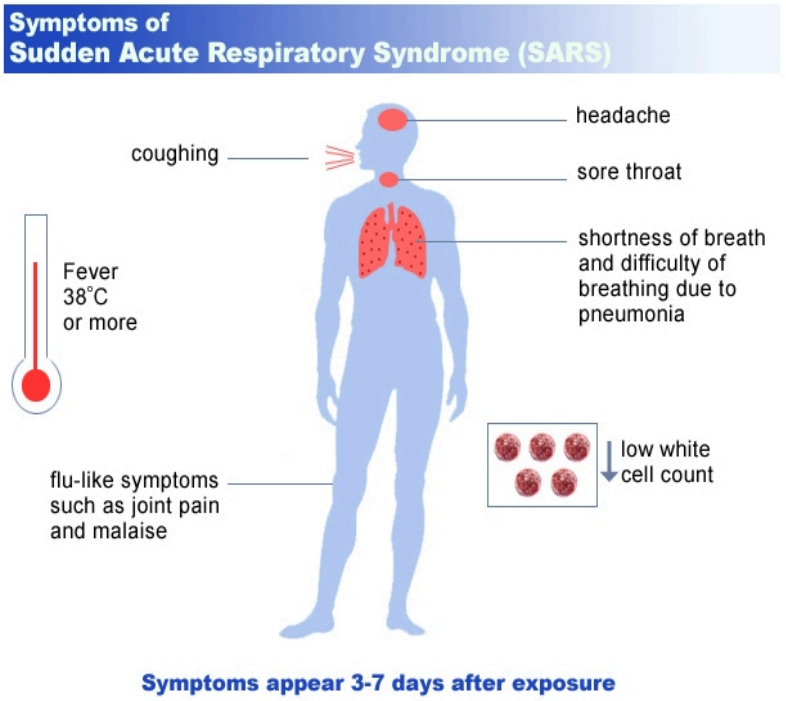


Transmission of SARS

- Close contact with large respiratory droplets
- Transmission by fomites possible
- Those severely ill more infectious (attack rate of >50% in some hospital staff)
- Infectivity increases during 2nd week of illness
- May remain infectious up to 10 days once afebrile

SARS: Symptoms

- Clinical course:
  - Short incubation period (2-3 days)
  - Intensive care if needed occurs about 2 weeks to a month after onset of symptoms
- There are generally 3 phases:
  - Week 1: cold-like symptoms, fever, myalgia, chill and a sore throat
  - Week 2, recurrence of fever. Onset of diarrhea, and oxygen desaturation
  - Only 20% of patients reach this phase, requires ventilatory support
  - The fatality rate is about 10% upon infection (average but burden higher for populations over 65 years old)
  - This information in unvaccinated/boosted



SARS: Diagnosis

- RT-PCR positive for SARS virion using validated methods on at least 2 different clinical specimens
- ELISA Antigen test
- Seroconversion (gold standard)

(negative antibody test on acute specimen followed by positive test on convalescent sera or > 4 rise in antibody titre between acute and convalescent sera)

Treatment for SARS

- No specific antiviral drugs available
- Vaccines are under development:
  - In China, a second-phase human trials of a SARS vaccine (inactivated)
    - Have safety concerns
  - The first U.S. SARS vaccine trial began at the NIH by using recombinant plasmid for S protein expression
  - Development of transgenic tomato & tobacco expressing SARS virion S protein
  - An experimental attenuated Vesicular Stomatitis Virus (VSV)-based vaccine

Prevention of SARS



Wearing N95 mask was one of the main public preventive measures against SARS

- Close contacts of a probable or suspect case of SARS are at high risk of becoming a case (should be isolated)
- Travelers from areas with recent local transmission of SARS should be given advice about whom to contact, and what to do if they develop a fever within their potential incubation period
- Health departments should assign staff to undertake contact tracing and daily review of contacts in the community.
- Recovered patients quarantined at home for at least 14 days after discharged from hospital

Transmission of SARS

- Close contact with large respiratory droplets or airborne
- Transmission by fomites possible but rare
- Infectivity starts during asymptomatic phase and well into 2nd week of illness or if person is shedding virus
- May remain infectious up to 10 days once afebrile
- *Differs between different variants*

Challenges?

- No treatment or last resort treatments
- Rare
- Not-equipped
- Monitoring emerging antimicrobial resistance
- Decline in antimicrobial susceptibility testing (proliferation Nucleic amplification testing bypass culture)
- Pandrug resistance (PDR)

Summary

- Defined emerging infectious disease
- Causes that lead to the emergence of these diseases and efforts to combat that.
- Viral and bacterial pathogens, their properties, pathogenesis, geographical distribution, etc.
- Challenges emerging infectious diseases present to the medical community.